

Notes: Farhi and Maggiori (QJE, 2018)

A Model of the International Monetary System

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1 Big picture

1.1 Question

Farhi and Maggiori build a tractable theory of the **international monetary system (IMS)**. In their framework, the IMS is organized around three linked objects: the world supply and demand for reserve assets, the exchange-rate regime, and the institutions or arrangements that determine who can safely issue those reserve assets.

The central question is why reserve-currency systems can look stable for long periods and then suddenly become fragile, and whether a world with several reserve issuers is necessarily better than a hegemonic dollar- or sterling-style system.

The paper uses one common framework to reinterpret several classic debates: Triffin's dilemma, the Despres–Kindleberger–Salant view of the United States as a world banker, Nurkse's warning about multipolar instability, and Keynes's argument that reserve scarcity under gold or at the zero lower bound can be recessionary.

1.2 Setting

The model starts with a single reserve issuer, called the **hegemon**, and a rest of the world, called **RoW**. RoW wants safe reserve assets because it is risk averse and faces risky investment opportunities. The hegemon can issue nominal debt in its own currency. This debt is useful to RoW only if it is expected to remain safe in bad states.

The crucial friction is **limited commitment**. After a bad state, the hegemon may choose to devalue its currency. A devaluation reduces the real value of its nominal liabilities, so it acts like a partial default on foreign reserve holders. But devaluation is costly because it damages reputation or future reserve-currency rents.

Interpretation. The model treats reserve-currency status as a form of market power. The hegemon sells safety to the rest of the world and collects a safety premium. The problem is that too much issuance can make the promise of safety noncredible.

1.3 Core challenge

The key economic tension is that reserve assets are both useful and fragile. The hegemon can earn monopoly rents by issuing reserve debt, but larger issuance increases the temptation to devalue in bad states. This creates an endogenous boundary between three regions:

- **Safety zone:** debt is low enough that the hegemon never devalues, regardless of investor beliefs.
- **Instability zone:** debt is intermediate, so both a safe equilibrium and a crisis equilibrium are possible.
- **Collapse zone:** debt is high enough that the hegemon devalues in bad states regardless of investor beliefs.

1.4 Main contribution

The paper’s contribution is not an empirical estimate. It is a theoretical synthesis. It provides a model that can simultaneously explain the supply of reserve assets, reserve-currency rents, confidence crises, exchange-rate regimes, and the welfare effects of competition among reserve issuers.

The model combines ideas from sovereign debt crises, monopoly and oligopoly theory, and Keynesian macroeconomics. The result is a simple but powerful framework in which reserve-currency status is valuable, but the private incentives of reserve issuers need not coincide with world welfare.

1.5 Main result

The paper delivers five central results.

First, under full commitment, a hegemon issues safe debt and earns monopoly rents from the safety premium. This is the model’s formal version of the **exorbitant privilege**.

Second, under limited commitment, the hegemon faces a **Triffin dilemma**. If world demand for reserves is high relative to the hegemon’s safe debt capacity, the hegemon must choose between restricting issuance to remain safe or expanding issuance and risking a confidence crisis.

Third, the hegemon may **underissue or overissue** reserve assets relative to social welfare. Underissuance is the standard monopoly distortion. Overissuance can arise because the hegemon does not internalize the inframarginal welfare loss suffered by RoW if a crisis destroys the safety value of outstanding reserve assets.

Fourth, reserve scarcity under a **gold-exchange standard** or under **floating exchange rates at the zero lower bound (ZLB)** can be recessionary. In those regimes, reserve scarcity cannot be fully absorbed by interest-rate adjustment, so output falls.

Fifth, a **multipolar** reserve system is not automatically better. With full commitment, more issuers raise total reserve supply and reduce monopoly rents. With limited commitment, a small number of competing reserve issuers can worsen coordination problems and make the IMS less stable, echoing Nurkse’s warning.

1.6 Economic takeaway

The paper bridges two views of the IMS. The “world banker” view is correct in the sense that a reserve issuer intermediates between global demand for safety and risky investment opportunities. But the Triffin view is also correct because banking is fragile: a large reserve issuer can become vulnerable to self-fulfilling runs. The model’s core message is that reserve-currency privilege is valuable precisely because safety is scarce, but scarcity and privilege also create incentives that can destabilize the system.

2 Historical facts and model objects

2.1 Historical motivation

The historical evidence motivates the model in two ways. First, reserve systems have often been concentrated around one or two currencies. Sterling dominated the classical gold standard; sterling and the dollar competed during the interwar gold-exchange standard; the dollar dominated Bretton Woods and the modern floating-rate system.

Second, reserve systems have experienced abrupt collapses after long periods of stability. Britain devalued sterling and left gold in 1931. The United States devalued the dollar in 1933 and again during the 1971–1973 collapse of Bretton Woods. The model interprets these episodes as confidence crises rooted in limited commitment.

2.2 The two agents

There are two classes of agents.

- **Hegemon.** A risk-neutral reserve issuer that can issue nominal debt denominated in its own currency.
- **Rest of the world (RoW).** A competitive fringe of international investors with mean-variance preferences, who demand safe reserve assets to hedge risk.

The hegemon is a monopolist in the baseline model. Later, the paper introduces multiple reserve issuers and analyzes Cournot competition in reserve-asset supply.

2.3 Preferences

RoW has mean-variance utility over period-1 consumption:

$$U^*(C_1^*) = E[C_1^*] - \gamma \text{Var}(C_1^*).$$

Interpretation. RoW values expected consumption but dislikes consumption risk. The parameter γ measures risk aversion. A larger γ means stronger demand for safe assets.

The hegemon is risk neutral:

$$U(C_0, C_1) = C_0 + \delta E[C_1], \quad \delta = \frac{1}{R^r}.$$

Interpretation. The hegemon is willing to shift resources between periods at the expected return on the risky real asset. Its motive for issuing reserve debt is therefore not risk sharing for itself, but the monopoly rent from selling safety.

2.4 Assets and states

There are two periods and two states at date 1. The low state L is a disaster and occurs with probability λ . The risky real asset has return $R_H^r > 1$ in the high state and $R_L^r < 1$ in the low state.

Let

$$\bar{R}^r = E[R^r], \quad \sigma^2 = Var(R^r).$$

The hegemon issues nominal debt. Its real repayment depends on the exchange rate. Let e denote the proportional exchange-rate change. A lower e is a devaluation. The paper normalizes the no-devaluation value to $e_H = 1$ and allows a devaluation $e_L < 1$ in bad states.

2.5 Devaluation as partial default

If the hegemon devalues, the real repayment on its nominal liabilities falls. This is equivalent to a partial default in the baseline real model. The hegemon pays a cost

$$\tau(1 - e_L),$$

where τ captures the reputation, institutional, or future-rent cost of devaluation.

Interpretation. The parameter τ is the model's commitment capacity. A larger τ means the hegemon is more reluctant to devalue, so it can safely issue more reserve debt. A smaller τ means reserve-currency status is more fragile.

2.6 Calvo timing and self-fulfilling crises

The timing follows the Calvo logic of self-fulfilling debt crises.

1. At $t = 0^-$, the hegemon chooses how much reserve debt b to issue.
2. At $t = 0^+$, a sunspot determines investor expectations. If expectations are favorable, debt is priced as safe; if expectations are unfavorable, debt is priced as risky.
3. At $t = 1$, the disaster state may occur. Conditional on the state, the hegemon decides whether to devalue.

Interpretation. The real quantity of nominal debt is chosen before the interest rate is determined. This timing makes beliefs important: if investors demand a high rate because they expect devaluation, the fiscal burden of debt rises, and devaluation can become optimal ex post.

3 Models and equilibrium specifications

3.1 RoW demand for safe reserve assets

If hegemon debt is expected to be safe, RoW's portfolio problem gives a linear inverse demand curve:

$$R_s(b) = \bar{R}^r - 2\gamma(w^* - b)\sigma^2.$$

Interpretation. $R_s(b)$ is the interest rate on safe hegemon debt. As the hegemon issues more safe debt, RoW holds less of the risky asset, so the marginal value of additional safety falls and the safe interest rate rises. The safety premium is

$$\bar{R}^r - R_s(b) = 2\gamma(w^* - b)\sigma^2.$$

This premium is higher when RoW wealth is larger, risk aversion is higher, or the risky asset is more volatile.

If hegemon debt is expected to be risky, it is a perfect substitute for the risky real asset. Then the interest rate must compensate investors for risk:

$$R = R_H^r.$$

3.2 The hegemon's issuance problem

The hegemon chooses issuance to maximize the expected transfer from RoW, net of expected devaluation costs:

$$\max_{b \geq 0} \{b(\bar{R}^r - E_-[R(b, \omega)e(b, \omega)]) - E_-[\tau(1 - e(b, \omega))]\}.$$

Interpretation. The hegemon earns a rent when it borrows at a safe rate below the expected risky return. But more issuance may increase the probability that its debt becomes risky and that the hegemon must pay the devaluation cost.

3.3 Full-commitment benchmark

Under full commitment, the hegemon can credibly promise never to devalue. Its problem becomes

$$\max_{b \geq 0} V^{FC}(b) = b(\bar{R}^r - R_s(b)).$$

Using the safe demand curve, the first-order condition gives

$$b^{FC} = \frac{1}{2}w^*, \quad R_s(b^{FC}) = \bar{R}^r - \gamma\sigma^2w^*.$$

The monopoly rent, or exorbitant privilege, is

$$b^{FC}(\bar{R}^r - R_s(b^{FC})) = \frac{1}{2}\gamma\sigma^2(w^*)^2.$$

Interpretation. Under full commitment, the hegemon acts like a monopolist selling safe assets. It restricts quantity below the competitive level w^* , charges a safety premium, and receives a transfer from RoW.

3.4 Perfect competition benchmark

With perfect competition and full commitment, safe returns are bid up until there is no safety premium:

$$R_s(b) = \bar{R}^r, \quad b = w^*.$$

Interpretation. Competition eliminates the reserve-currency rent and provides full insurance to RoW. This benchmark explains why a multipolar system can look attractive: it increases reserve supply and reduces the monopoly distortion.

3.5 Limited commitment and the devaluation condition

In a disaster, the hegemon devalues if the gain from reducing real repayment exceeds the cost of devaluation. The decision rule is

$$bR(1 - e_L) > \tau(1 - e_L),$$

or equivalently

$$bR > \tau.$$

Interpretation. The product bR is the real repayment burden. A high debt quantity or a high crisis interest rate makes devaluation more tempting. This is the core limited-commitment friction.

3.6 The three zones of the IMS

The devaluation condition generates three zones.

The lower boundary of the safety zone is characterized by the risky-rate threshold

$$\underline{b} = \frac{\tau}{R_H^r}.$$

If $b \leq \underline{b}$, even pessimistic pricing does not make devaluation attractive, so debt is safe.

The upper boundary of the instability zone is

$$\bar{b} = \frac{-\bar{R}^r + 2w^*\gamma\sigma^2 + \sqrt{(\bar{R}^r - 2w^*\gamma\sigma^2)^2 + 8\gamma\sigma^2\tau}}{4\gamma\sigma^2}.$$

If $b > \bar{b}$, even optimistic safe pricing leaves the hegemon with too much repayment burden, so devaluation occurs in the disaster state.

- $b \in [0, \underline{b}]$: unique safe equilibrium.
- $b \in (\underline{b}, \bar{b}]$: multiple equilibria; safe or collapse depending on expectations.
- $b \in (\bar{b}, w^*]$: unique collapse equilibrium.

Interpretation. The instability zone is the formal Triffin region. Reserve assets are safe only as long as investors believe they are safe. Pessimistic beliefs raise the interest burden, validate devaluation, and destroy the safety premium.

3.7 Sunspot selection and limited-commitment value

The paper uses a simple sunspot selection rule. Let $\alpha(b)$ be the probability of the collapse equilibrium:

$$\alpha(b) = \begin{cases} 0, & b \in [0, \underline{b}], \\ \alpha, & b \in (\underline{b}, \bar{b}], \\ 1, & b \in (\bar{b}, w^*]. \end{cases}$$

The limited-commitment value is

$$V(b) = (1 - \alpha(b))V^{FC}(b) - \alpha(b)\lambda\tau(1 - e_L).$$

Interpretation. Issuing in the instability zone preserves monopoly rents only with probability $1 - \alpha$ and exposes the hegemon to expected devaluation cost with probability $\alpha\lambda$. The hegemon compares a low-issuance safe option with a high-issuance risky option.

4 Mechanisms and economic interpretation

4.1 The Triffin dilemma

The Triffin dilemma arises when full-commitment issuance exceeds safe debt capacity:

$$b^{FC} > \underline{b}.$$

Equivalently, world reserve demand is large relative to the hegemon's commitment capacity. The hegemon then faces a choice:

- issue only $\underline{b}$ and remain safely within the safety zone;
- issue more, move into the instability zone, and risk a confidence crisis.

Interpretation. The dilemma is not simply that the hegemon must run external deficits. It is that the reserve issuer's gross external liabilities eventually become large enough that the promise not to devalue becomes vulnerable to beliefs.

4.2 Exorbitant privilege

The hegemon earns an exorbitant privilege because safe reserve assets trade at a lower return than risky assets. In the model, this privilege is a monopoly rent:

$$\text{Privilege} = b(\bar{R}^r - R_s(b)).$$

Interpretation. The privilege is not free money generated by accounting. It is payment for scarce safety. The hegemon supplies a special asset that RoW values because it hedges risk and provides liquidity-like benefits.

4.3 Gross liabilities matter

The model highlights **gross** external safe liabilities, not simply the hegemon's net international investment position. A reserve issuer can look solvent on a net basis while still owing large safe-currency liabilities to foreigners. Those gross liabilities are the claims that may be inflated away by devaluation.

Interpretation. The relevant fragility measure is the scale of reserve promises outstanding. A large world banker may hold risky assets on one side of its balance sheet and issue safe liabilities on the other; the risk is a run on the liabilities.

4.4 Welfare: underissuance and overissuance

The welfare analysis adds RoW expected utility to the hegemon's monopoly rents. A standard monopoly intuition suggests underissuance: the hegemon restricts quantity to keep the safety premium high.

But the paper shows that overissuance is also possible. The reason is that issuance changes not only quantity but also risk. If high issuance moves the economy into the instability zone, a crisis destroys safety on the whole inframarginal stock of reserve assets. The hegemon does not internalize RoW's loss from this destruction of safety.

4.5 Convex reserve demand and overissuance

To study welfare, the paper allows the reserve demand curve to be convex in the safety premium. A tractable version is

$$R_s(b) = \bar{R}^r - 2\gamma(w^* - b)\sigma^2 - 2\gamma_L(b - \underline{b})\mathbf{1}_{\{b \leq \underline{b}\}}.$$

The welfare of RoW can be represented as the area under the inverse demand curve:

$$V_{RoW}(b) = V_{RoW}(0) + (1 - \alpha(b)) \int_{R_s(0)}^{R_s(b)} b(\tilde{R}_s) d\tilde{R}_s.$$

Interpretation. If the first units of safe assets are especially valuable, then destroying the safety of the inframarginal stock is very costly for RoW. In that case, a hegemon may issue too much from a social point of view.

4.6 Proposition 3 in words

The welfare result can be summarized as follows.

- With a linear demand curve, the hegemon underissues relative to social welfare, as in a standard monopoly problem.
- With sufficiently convex demand and low enough devaluation cost, the hegemon may overissue.
- Misalignment is most likely for intermediate crisis probabilities. If crisis probability is very low, both the hegemon and RoW like high issuance. If it is very high, both prefer safety.

Interpretation. The paper's welfare insight is that reserve supply is a quantity-quality problem. More issuance increases quantity but can lower quality by making reserve assets fragile.

5 Exchange-rate regimes and multipolarity

5.1 Floating exchange rates and the ZLB

The paper extends the baseline model by adding production and nominal rigidities. RoW demand for safe assets becomes

$$R_s(b; \ell) = \bar{R}^r - 2\gamma\sigma^2(w_e^* + \ell - b),$$

where w_e^* is endowment wealth and ℓ is labor income/output.

If the economy is away from the ZLB, monetary policy can adjust the nominal interest rate to maintain full employment. If the ZLB binds, the nominal interest rate cannot fall far enough. The ZLB threshold is

$$b_{ZLB} = \frac{1 - \bar{R}^r + 2\gamma\sigma^2(w_e^* + \bar{L})}{2\gamma\sigma^2}.$$

If $b < b_{ZLB}$, safe assets are scarce at full employment. Output must fall until reserve demand is low enough to clear the market:

$$R_s(b; \ell) = 1.$$

Interpretation. At the ZLB, reserve scarcity becomes a Keynesian aggregate-demand problem. Issuing more safe assets raises aggregate demand because it satisfies the demand for safety that interest-rate policy cannot accommodate.

5.2 Gold-exchange standard

Under a gold-exchange standard, monetary policy is tied to a fixed nominal price of gold. The interest rate is pinned down by the gold parity rather than by domestic stabilization needs. If reserve assets are scarce at full employment, output falls.

The paper shows that a gold-exchange standard can intensify the Triffin dilemma. Because the demand for reserve assets becomes very elastic, the hegemon has stronger incentives to issue. But if a confidence crisis occurs, the effective stock of reserves collapses, and the recession becomes more severe.

Interpretation. This mechanism rationalizes Keynes's warning that returning to gold at tight parities could be recessionary. It also explains why monetary reserves were used to supplement gold in the interwar gold-exchange standard, and why the collapse of reserve confidence was so damaging.

5.3 Expenditure-switching motive to devalue

In the baseline model, the incentive to devalue is fiscal: reduce real debt repayment. With nominal rigidities and nontraded goods, there is also a macroeconomic motive: devaluation can stimulate domestic output through expenditure switching.

The effective devaluation cost becomes

$$\tilde{\tau} = \tau - \frac{\nu(C_{NT,1}(e_L)) - \nu(C_{NT,1}(1))}{1 - e_L} < \tau.$$

Interpretation. A recession makes devaluation more attractive. Thus confidence crises are more likely when the hegemon also has domestic stabilization reasons to weaken its currency. This helps interpret sterling in 1931, the dollar in 1933, and the dollar in 1971–1973.

5.4 Multipolar reserve supply under full commitment

The multipolar model has n symmetric reserve issuers competing in quantities. Under full commitment, issuer i faces

$$R_s(b_i, b_{-i}) = \bar{R}^r - 2\gamma\sigma^2(w^* - b_i - b_{-i}).$$

The Cournot equilibrium is symmetric:

$$b_i^{FC} = \frac{1}{n+1}w^*, \quad B_n^{FC} = \frac{n}{n+1}w^*,$$

with

$$R_s(B_n^{FC}) = \bar{R}^r - \frac{2}{n+1}\gamma\sigma^2w^*.$$

Interpretation. More reserve issuers raise total safe-asset supply and reduce the safety premium. In the limit as $n \rightarrow \infty$, the economy approaches the competitive full-insurance benchmark.

5.5 Nurkse instability under limited commitment

With limited commitment, competition has a darker side. Multiple reserve currencies create coordination problems: investors may switch away from one reserve asset and into another, making the abandoned issuer more likely to devalue.

The paper shows that the benefits of multipolarity may be **U-shaped** in the number of issuers. With many issuers, each one issues little and remains safe. With only a few issuers, competition can create instability without fully eliminating monopoly power.

Interpretation. This result reconciles two views. Eichengreen-style optimism is right under full commitment or with many issuers. Nurkse-style pessimism is relevant when there are only a few large reserve issuers and investor coordination is fragile.

5.6 Other extensions

The paper also shows how the framework can accommodate several additional features.

- **Endogenous reputation:** devaluation can reduce future reserve-currency rents.
- **Private reserve issuance:** private safe assets denominated in the reserve currency can expand reserve supply but still depend on sovereign credibility.
- **Fiscal capacity:** countries with stronger fiscal capacity can issue more reserves.
- **Currency of pricing:** a currency used in goods pricing may issue safer nominal debt.
- **Liquidity and network effects:** deep markets can reinforce hegemonic status.
- **Endogenous entry:** reserve-currency status can resemble a natural monopoly because reputation is costly to acquire.
- **Global safety nets:** swap lines, IMF liquidity, and reserve-sharing arrangements can reduce reserve demand and mitigate the Triffin dilemma.

6 Figures and propositions

6.1 Figure I: History of the international monetary system

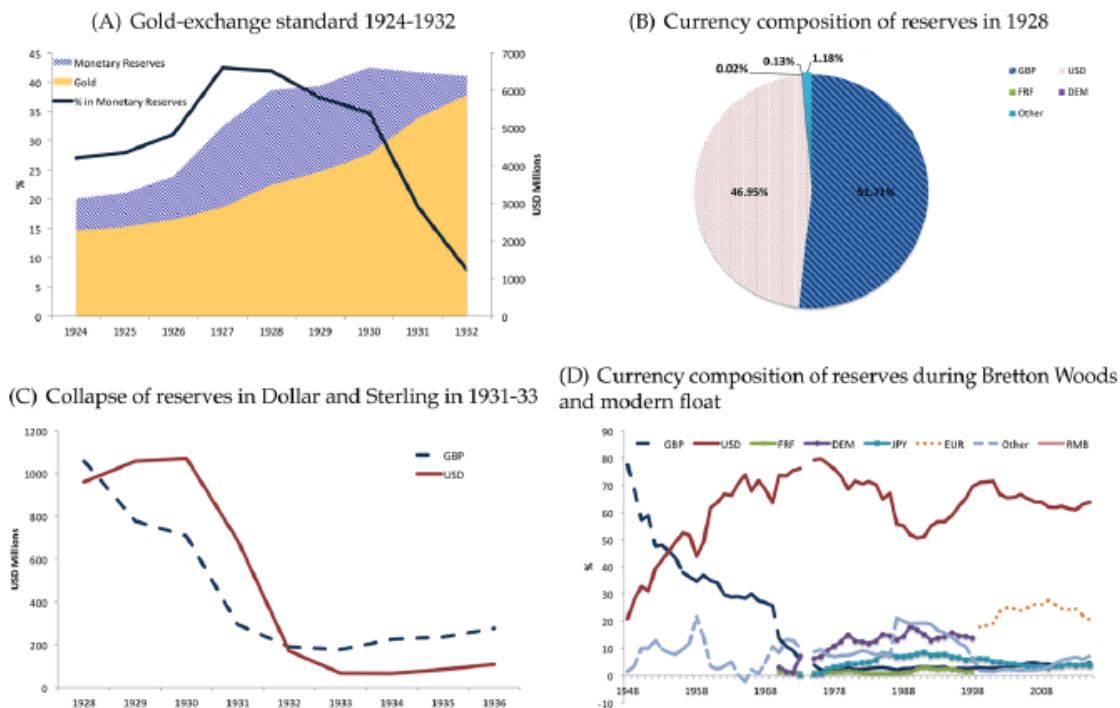


Figure 1: Historical reserve patterns in Farhi and Maggiori.

Read: Figure I shows the expansion and collapse of the interwar gold-exchange standard, the dollar-sterling composition of reserves in 1928, the collapse of dollar and sterling reserves in 1931–1933, and the currency composition of reserves under Bretton Woods and the modern floating system.

Economic message: The figure motivates the model. Reserve systems can be concentrated, can shift across currencies, and can collapse abruptly. The historical patterns line up with the paper’s emphasis on hegemonic reserve status, multipolar instability, and confidence crises.

6.2 Figure II: Timing of the baseline model

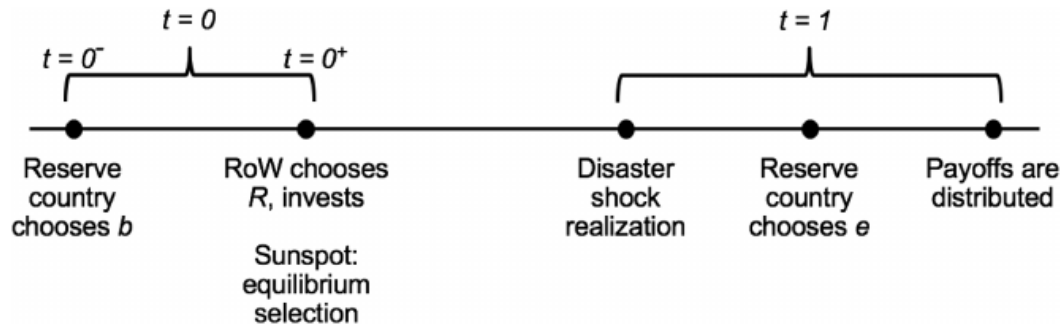


FIGURE II

Figure 2: Timing in the one-period model.

Read: The hegemon first chooses issuance b . Investors then price the debt and choose portfolios after a sunspot selects beliefs. Finally, the disaster state may occur and the hegemon chooses whether to devalue.

Economic message: The timing is essential. Debt quantity is fixed before the interest rate is determined. This creates the self-fulfilling crisis logic: pessimistic expectations can raise the repayment burden and make devaluation optimal.

6.3 Figure III: Hegemon optimal debt issuance

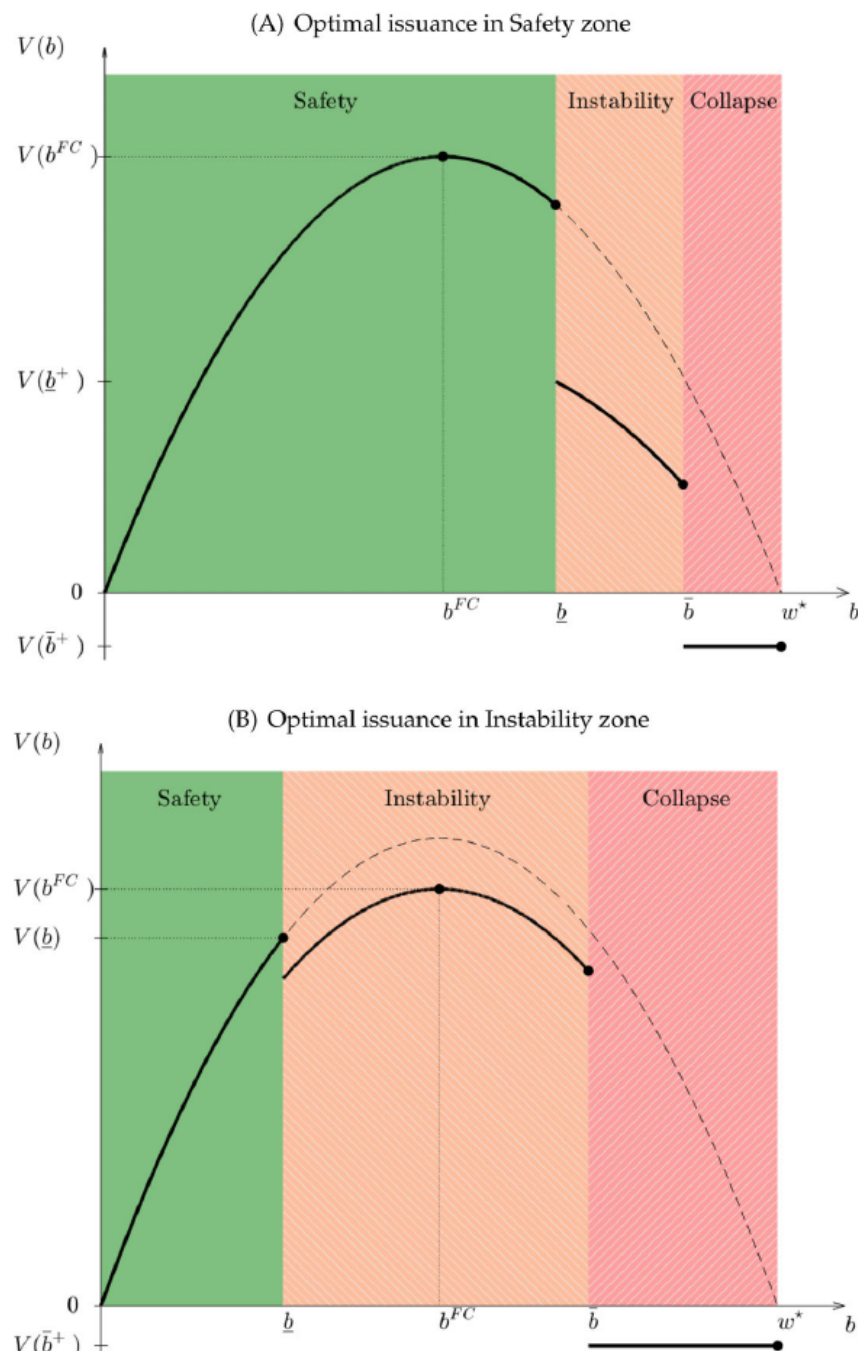


Figure 3: Safety, instability, and collapse zones.

Read: Panel A shows a case where full-commitment issuance lies in the safety zone. Panel B shows a case where the hegemon chooses issuance in the instability zone despite the possibility of collapse.

Economic message: This is the graphical version of the Triffin dilemma. When world reserve

demand is high or commitment is weak, the hegemon may stretch beyond safe debt capacity to preserve monopoly rents. The private gain from issuing more must be weighed against the possibility of a confidence crisis.

6.4 Figure IV: Welfare consequences of the Triffin dilemma

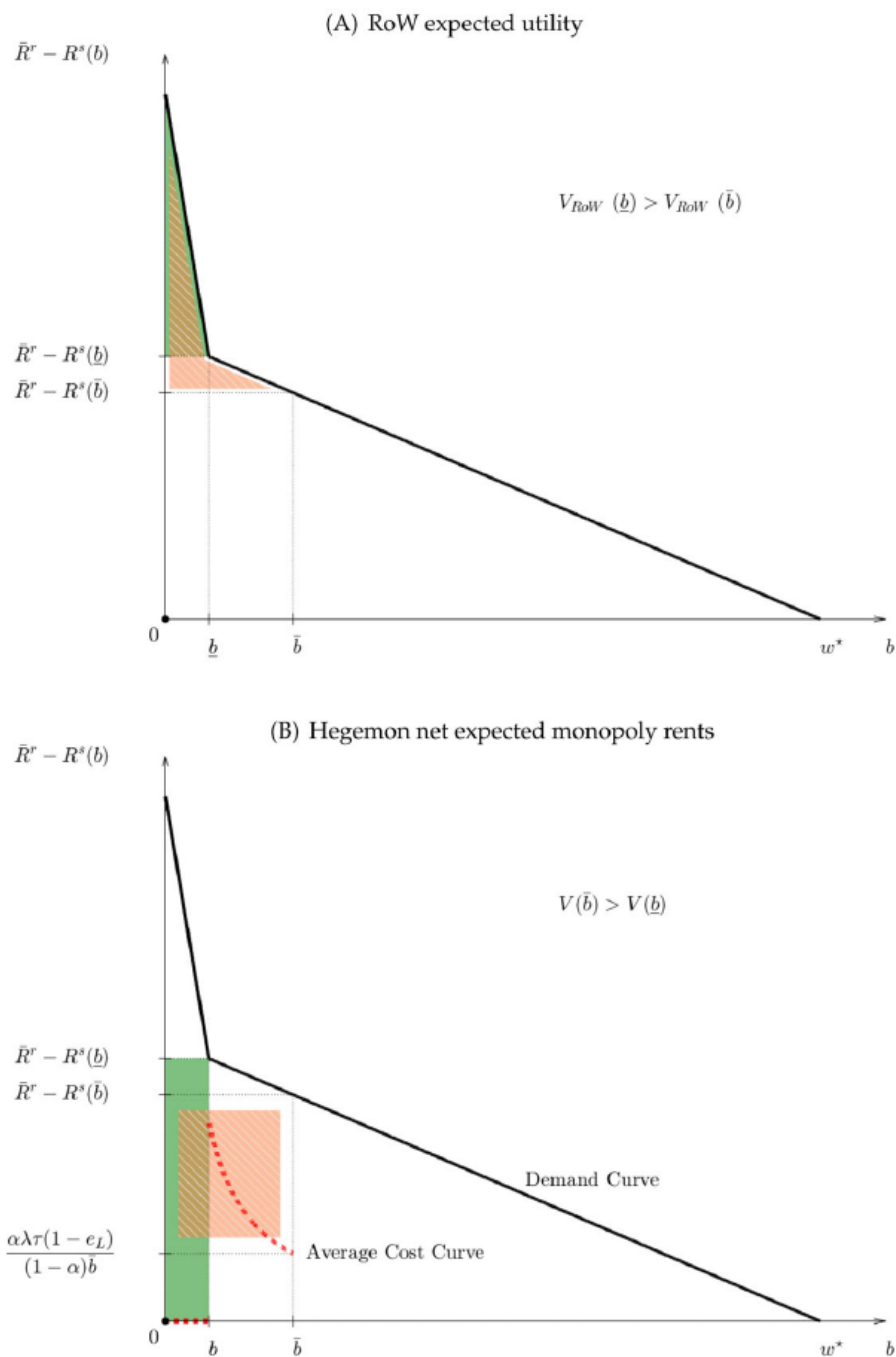


Figure 4: RoW welfare and hegemon rents under safe versus risky issuance.

Read: The top panel represents RoW expected utility as areas under the demand curve. The bottom panel represents the hegemon's expected monopoly rents. The safe option and instability-zone option can be ranked differently by RoW and by the hegemon.

Economic message: The figure explains overissuance. The hegemon considers its own rent and devaluation cost, but does not internalize how a crisis destroys the inframarginal safety surplus of RoW. When the demand for safe assets is convex, this missing welfare term can be large.

6.5 Figure V: Overissuance versus underissuance

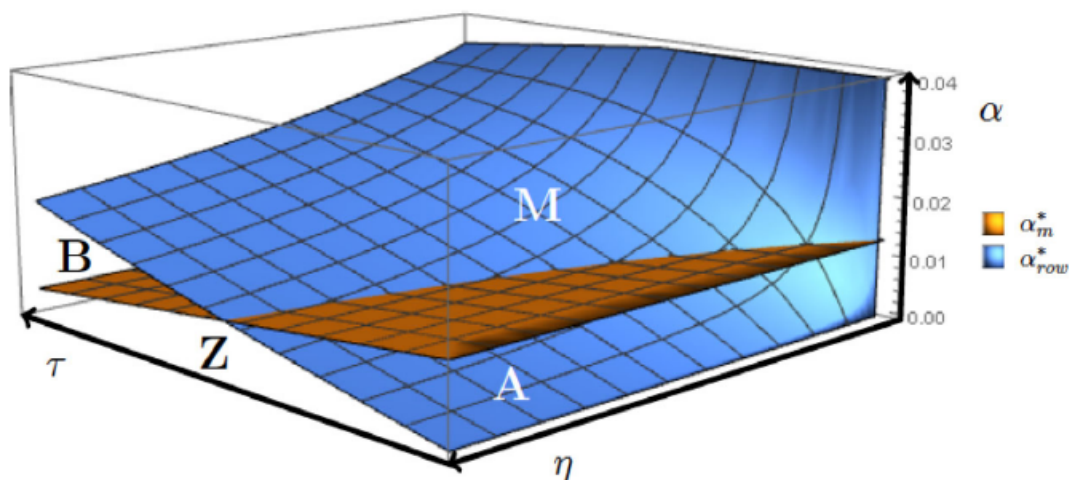


Figure 5: Parameter regions for overissuance and underissuance.

Read: Figure V maps the regions in which the hegemon and RoW prefer different issuance choices. The key parameters are the commitment cost, the convexity of demand, and the crisis probability.

Economic message: The figure shows that neither overissuance nor underissuance is universal. Underissuance is more likely with nearly linear demand and strong commitment. Overissuance is more likely with convex demand, weak commitment, and intermediate crisis probabilities.

6.6 Figure VI: Gold-exchange standard and ZLB

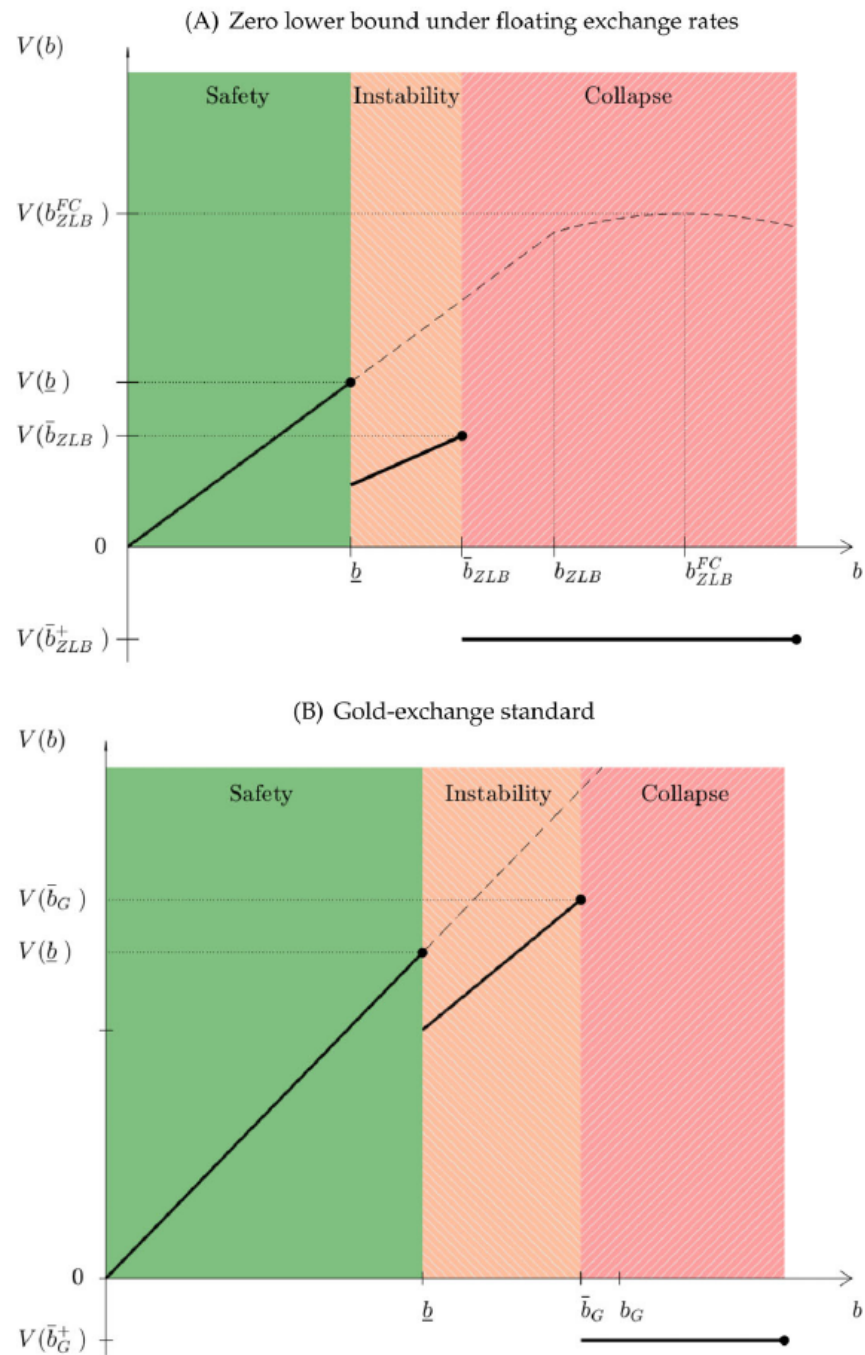


Figure 6: Reserve issuance under floating rates at the ZLB and under a gold-exchange standard.

Read: Panel A shows the ZLB under floating exchange rates. Panel B shows the gold-exchange standard. In both cases, the hegemon's value function changes because reserve scarcity cannot be fully offset by interest-rate adjustment.

Economic message: The figure connects Triffin fragility with Keynesian demand management. Safe-asset issuance can support output when reserves are scarce, but higher issuance can also increase crisis risk. The same debt that stabilizes demand today may destabilize the reserve system tomorrow.

6.7 Proposition map

- **Proposition 1:** Under full commitment, the hegemon issues safe debt, charges a safety premium, and earns monopoly rents.
- **Proposition 2:** Under limited commitment, the hegemon may choose safe low issuance or risky high issuance; this formalizes the Triffin dilemma.
- **Proposition 3:** The hegemon may underissue or overissue depending on demand convexity, commitment, and crisis probability.
- **Proposition 4:** At the ZLB, reserve scarcity can reduce output; issuance may stimulate demand but increase fragility.
- **Proposition 5:** Under a gold-exchange standard, reserve scarcity and confidence crises can create or deepen recessions.

7 Short summary

- The paper models the IMS as a market for scarce safe reserve assets.
- A hegemon can issue reserve debt and collect a safety premium because RoW values safety.
- Under full commitment, the hegemon behaves like a monopolist and earns an exorbitant privilege.
- Under limited commitment, high issuance can make devaluation tempting in bad states.
- The economy has a safety zone, an instability zone, and a collapse zone.
- The Triffin dilemma occurs when world reserve demand exceeds the hegemon's safe debt capacity.
- The hegemon may underissue because of monopoly power or overissue because it does not internalize crisis losses to RoW.
- At the ZLB or under a gold-exchange standard, reserve scarcity can be recessionary.
- A multipolar system raises reserve supply under full commitment, but limited commitment can create Nurkse-style instability.
- The paper's main lesson is that reserve-currency status is both a privilege and a source of systemic fragility.

8 Conclusion

Farhi and Maggiori provide a unified model of reserve-currency systems. The core mechanism is simple: the world demands safe assets; only a few countries can credibly issue them; and issuing too many safe promises can make the promise of safety fragile.

The model explains why a hegemon earns an exorbitant privilege, why the Triffin dilemma arises, why gold and ZLB regimes can turn reserve scarcity into recession, and why multipolarity is not automatically stabilizing. It also clarifies why the IMS can survive for long stretches and then collapse suddenly: investor beliefs matter once reserve liabilities are large enough to make devaluation tempting.

The final takeaway is that the IMS is not just about exchange-rate arrangements or balance-of-payments accounting. It is a global market for safety, and that market is shaped by monopoly power, commitment, and coordination. The same scarcity that makes reserve assets valuable also makes the system fragile.